Highlights

Verified coupled simulation and uncertainty analysis of floating photovoltaic lifetime performance in humid and saline environments

Subrahmanyam Tanala

- Coupled hourly simulation links FPV cooling, humidity, salt soiling, PID, corrosion
- Sub-models agree with pvlib to 3e-12; Monte Carlo statistics converge at 1,000 runs
- Freshwater FPV lifetime gain is conditional: positive in 75% of parameter samples
- Saline FPV shows a lifetime energy penalty across most of the parameter space
- Sobol indices rank float heat loss, land dust and salt deposition as key drivers

Verified coupled simulation and uncertainty analysis of floating photovoltaic lifetime performance in humid and saline environments

Subrahmanyam Tanala*

[a]*Department of Electrical and Electronics Engineering, Anil Neerukonda Institute of Technology and Sciences (ANITS), Visakhapatnam, 531162, Andhra Pradesh, India*

Abstract

Floating photovoltaic (FPV) plants are expected to outperform land-based plants because water cools the modules, but the same water raises humidity and, at saline sites, airborne salt, which accelerate soiling and degradation. This paper presents a simulation framework in which one over-water microclimate drives module temperature, dew, a dust and sea-salt soiling balance with NaCl deliquescence, and four degradation mechanisms (intrinsic, Peck temperature–humidity with lagged encapsulant moisture, salt-assisted potential-induced degradation and chloride corrosion), feeding 25-year energy and levelized cost. The established sub-models are verified code-to-code against pvlib to within 3×10^{-12}, and every component is classified by its evidential basis. Weather variability and parameter uncertainty (28 parameters) are separated using a convergence-checked Monte Carlo and Sobol variance decomposition. For a hot, humid, coastal site in India, the baseline 25-year energy relative to land PV is +2.1% for freshwater and −4.3% for saline FPV. Under parameter uncertainty the freshwater gain is conditional (median +1.5%, positive in 75% of samples), whereas the saline penalty is robust (median −5.1%). Float heat-loss coefficients, land dust load and salt deposition explain most of the variance. The results are model-based scenario predictions, and a field-validation protocol is proposed.

Keywords: Floating photovoltaics, Coupled simulation, Verification and validation, Uncertainty quantification, Sobol sensitivity analysis, Module degradation

*Corresponding author.
Email address: subrahmanyam.eee@anits.edu.in (Subrahmanyam Tanala)

1. Introduction

Floating photovoltaic (FPV) systems place PV arrays on pontoons over reservoirs, lakes, lagoons and sheltered coastal waters. They avoid competition for land, reduce evaporation from water-supply reservoirs and can share grid infrastructure with hydropower plants [1, 2, 3, 4, 5]. Many candidate sites in South and South-East Asia are hot, humid and exposed to marine aerosol [6]. In these regions, the decision between floating and land-based PV depends on lifetime energy, not on first-year yield alone.

From a modelling perspective, this decision raises three difficulties. First, the mechanisms interact through a shared microclimate. The water lowers module temperature, which reduces thermal losses and Arrhenius-type ageing [7, 8, 9]. It also raises humidity, which increases moisture ingress and PID [10, 11, 12, 13]. At saline sites, salt deposits add soiling, lower the surface resistance of the glass [14, 15] and drive corrosion [16]. Simulating each mechanism in isolation, with independent inputs, cannot resolve these trade-offs. Second, the time scales differ: optical and thermal processes act hourly, soiling daily, and degradation over decades. Third, many of the governing parameters (heat-loss coefficients of a given float design, the humidity increment over water, salt and chloride deposition, degradation reference rates) are uncertain or unmeasured at a candidate site. Measured FPV heat-loss coefficients, for example, range from values similar to open-rack land PV [17, 18, 19] to values much higher for well-ventilated designs [8, 20, 21].

PV simulation practice has addressed parts of this problem. Uncertainty in long-term yield prediction has been quantified by statistical simulation [22], and blind intercomparisons show that modelling choices and derates still cause errors of several percent [23, 24]. Climate-dependent degradation models change lifetime yield estimates substantially compared with fixed rates [25, 26]. Variance-based sensitivity analysis has been applied to PV degradation in a humid tropical climate [27]. For offshore and floating PV, degradation has been forecast from climatic stresses [28], and salinity has been added to calibrated climatic degradation models [29]. Among the studies reviewed here (Table 1), none was identified that couples the yield

Table 1: Scope of representative studies (✓ = addressed).

Study	FPV	Thermal	Humidity	Salt	PID	Corrosion	Lifetime	UQ	Sobol
Thevenard and Pelland [22]							✓	✓	
Dörenkämper et al. [8]	✓	✓							
Golroodbari and van Sark [9]	✓	✓							
Kaaya et al. [26]		✓	✓				✓		
Mannino et al. [28]	✓	✓	✓				✓		
Hameed et al. [25]		✓	✓				✓		
Ajmal et al. [29]	✓	✓	✓	✓			✓		
Ahmad et al. [30]	✓	✓		✓					
Yuniawan et al. [27]		✓					✓	✓	✓
This work	✓	✓	✓	✓	✓	✓	✓	✓	✓

benefits of FPV and its humidity-, salt- and PID-driven penalties within one microclimate, verifies the implementation, and separates weather from parameter uncertainty with variance-based sensitivity analysis.

This paper contributes a simulation framework and its use for a decision problem:

1. A multi-time-scale coupled model in which one over-water microclimate drives module temperature, dew, spectral response, soiling and four module-degradation mechanisms (Section 2).
2. A verification and validation (V&V) assessment following established simulation practice [31, 32]. It comprises code-to-code verification of the established sub-models against pvlib, a classification of every component by its evidential basis, a consistency check of the weather generator, and a field-validation protocol (Section 3).
3. A designed set of simulation experiments that separates weather variability from parameter uncertainty. It combines a convergence-checked Monte Carlo with Sobol variance decomposition [33, 34] and break-even analyses (Section 4).
4. An application to coastal land PV, freshwater FPV and saline FPV in a hot, humid Indian climate, covering energy, degradation, mitigation and cost (Section 5).

The degradation results describe the electrical degradation of the PV modules. Degradation of the floating structure (float and mooring fatigue, structural corrosion [35, 36], storm damage, bio-fouling) is outside the model.

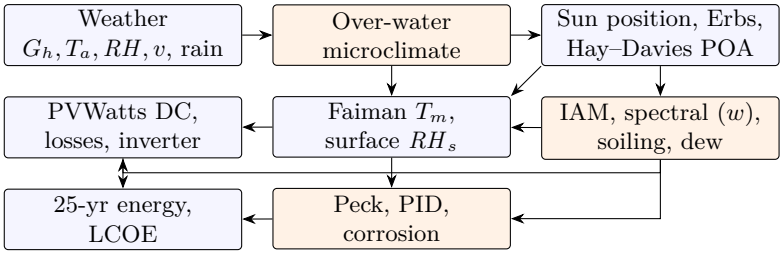


Figure 1: Coupled model chain. Shaded blocks depend explicitly on humidity or salinity.

Algorithm 1 Evaluation of one scenario

Require: hourly weather $\mathbf{W}$; scenario parameters $\boldsymbol{\theta}$

$\mathbf{W}' \leftarrow \text{microclimate}(\mathbf{W}; \alpha, \Delta T_d, k_v)$ ▷ Eq. (1)

sun position, E_0; Erbs decomposition; Hay–Davies G ▷ Eq. (3)

for each day d **do** ▷ soiling, Eq. (6)

    add dust and salt; apply rain cleaning (reduced after NaCl deliquescence) and scheduled cleaning

    $SR_d \leftarrow \max(SR_{\min}, 1 - k_d m_d - k_s m_s)$

end for

$T_m \leftarrow \text{Faiman}(G, T'_a, v')$; RH_s; dew state by hysteresis

$G_\text{eff} \leftarrow G \cdot \text{IAM} \cdot M(AM, w) \cdot SR \cdot (1 - \ell_\text{dew} \mathbb{1}_\text{wet})$

$P_\text{ac} \leftarrow \text{inverter}\big(\text{PVWatts}(G_\text{eff}, T_m) \prod_j (1 - \ell_j)\big) \cdot \text{availability}$

$RH_\text{e} \leftarrow$ lag of RH_s; evaluate $R_\text{TH}, R_\text{PID}, R_\text{C}$ ▷ Eqs. (7)–(11)

$R \leftarrow \sum$ mechanisms; $E_y \leftarrow E_1(1 - R/100)^{y-1}$; LCOE ▷ Eq. (12)

return year-1 losses, R by mechanism, 25-year energy, LCOE

2. Simulation model

2.1. Conceptual model and model chain

Fig. 1 shows the model chain. The model is a deterministic, discrete-time simulation with an hourly time step over one year (8,760 steps), a daily soiling sub-model, and an annual-rate degradation model extrapolated over the plant life. Algorithm 1 summarizes one scenario evaluation. The established irradiance and electrical sub-models follow their reference formulations, as implemented in pvlib [37, 38]; they are re-implemented in vectorized form and verified against pvlib in Section 3.

2.2. Weather input and over-water microclimate

The model accepts measured hourly weather. In the absence of a measured series for the reference site, a stochastic weather generator produces typical years from monthly normals. It draws daily rain occurrence and amounts, a first-order autoregressive daily clearness index and daily temperature anomalies. Hourly GHI is the Haurwitz clear-sky value [39] scaled by the clearness index. Air temperature follows an asymmetric diurnal cycle. The dew point is held constant over each day and shifted per month so that the monthly mean RH matches the normal. Wind speed includes a mean-preserving afternoon sea breeze. Over water, the air temperature is pulled towards the water temperature T_w (30-day running mean of air temperature plus 0.5 K), the dew point rises by ΔT_d, and the wind speed is scaled:

$$T'_a = T_a + \alpha\,(T_w - T_a), \qquad v' = k_v v, \qquad T'_d = \min(T_d + \Delta T_d,\, T'_a). \tag{1}$$

Relative humidity follows from the Magnus relation:

$$RH = 100\,\frac{e_s(T'_d)}{e_s(T'_a)}, \qquad e_s(T) = 6.112\exp\!\left(\frac{17.62\,T}{243.12 + T}\right)\ \text{hPa}. \tag{2}$$

2.3. Irradiance, incidence and spectrum

Solar position uses Spencer's series [40]. GHI is decomposed with the Erbs correlation [41] and transposed with the Hay–Davies model [42]:

$$G = G_b\cos\theta + G_d\left[A_i R_b + (1 - A_i)\tfrac{1+\cos\beta}{2}\right] + \rho\,G_h\tfrac{1-\cos\beta}{2}, \tag{3}$$

where θ is the angle of incidence, β the tilt, A_i the anisotropy index, R_b the beam ratio and ρ the albedo (0.07 for water, 0.20 for land). Beam reflection uses the ASHRAE modifier with $b_0 = 0.05$. The mono-Si spectral factor [43] is computed from the Kasten–Young air mass [44] and the precipitable water w [45]:

$$M = c_1 + c_2 AM + c_3 w + c_4\sqrt{AM} + c_5\sqrt{w} + c_6\,AM/\sqrt{w}. \tag{4}$$

2.4. Module temperature, surface humidity and dew

Module temperature follows the Faiman model [46], with a net long-wave loss $q_{\mathrm{sky}} = 40\,\mathrm{W/m^2}$ at night:

$$T_m = T'_a + \frac{G - q_{\mathrm{sky}}\,\mathbb{1}[G < 50]}{U_0 + U_1 v'}. \tag{5}$$

The RH of the air in contact with the glass is $RH_{\mathrm{s}} = 100\,e_s(T'_d)/e_s(T_m)$, so a hot module is effectively dry and a module colder than the dew point is wet. Dew forms when $T_m - T'_d < 1\,\mathrm{K}$ and persists until $T_m - T'_d > 4\,\mathrm{K}$. While dew is present, an assumed optical loss of 4% applies.

2.5. Dust and sea-salt soiling

Dust and salt surface densities (m_d, m_s in $\mathrm{g/m^2}$) follow a daily mass balance. Dust accumulates at r_d; salt at $r_s v/v_0$ with $v_0 = 3\,\mathrm{m/s}$. Rain above $2\,\mathrm{mm/day}$ removes a fraction ηf_β of each deposit ($\eta_d = 0.80$, $\eta_s = 0.95$, $f_\beta = \sin\beta/\sin 20^\circ$). When the daily maximum RH exceeds the NaCl deliquescence point of 75%, the recrystallized salt crust cements dust to the glass [30], and the dust-cleaning efficiency is reduced by a cementation factor (0.5). All plants are cleaned every 15 days. The soiling ratio is

$$SR = \max\left(SR_{\min},\, 1 - k_d m_d - k_s m_s\right). \tag{6}$$

The *salt deposition* r_s is the sea-salt mass retained on the glass. It is distinct from the *chloride deposition* D_{Cl} ($\mathrm{mg/(m^2\,d)}$), the chloride-ion deposition rate of the atmosphere in the ISO 9223 sense [16], which drives corrosion.

2.6. Electrical model

DC power follows the PVWatts model [47], $P_{\mathrm{dc}} = P_0(G_{\mathrm{eff}}/1000)[1 + \gamma(T_m - 25)]\prod_j(1 - \ell_j)$, with $\gamma = -0.0035/\mathrm{K}$ and ℓ_j the wiring, mismatch, wave-motion and light-induced degradation losses. The PVWatts inverter curve with clipping and an availability factor follow.

2.7. Module degradation model

The annual module degradation rate is the sum of four mechanisms, $R = R_0 + R_{\mathrm{TH}} + R_{\mathrm{PID}} + R_{\mathrm{C}}$, with $R_0 = 0.30\%/\mathrm{yr}$. Encapsulant moisture follows the surface humidity with a first-order lag ($\tau = 48\,\mathrm{h}$) [11]:

$$RH_{\mathrm{e}}(t) = RH_{\mathrm{e}}(t-1) + \kappa\,[RH_{\mathrm{s}}(t) - RH_{\mathrm{e}}(t-1)], \qquad \kappa = 1 - e^{-\Delta t/\tau}. \quad (7)$$

With $a_E(T_m) = \exp[-(E/k_B)(1/T_m - 1/T_{\mathrm{ref}})]$, the Peck rate [10] averaged over the year is

$$R_{\mathrm{TH}} = R_{\mathrm{TH}}^{\mathrm{ref}} \left\langle a_{E_a}(T_m) \left(\frac{\phi\, RH_{\mathrm{e}}}{RH_{\mathrm{ref}}} \right)^{n} \right\rangle, \quad (8)$$

with $E_a = 0.79\,\mathrm{eV}$, $n = 3$, $T_{\mathrm{ref}} = 25\,^{\circ}\mathrm{C}$, $RH_{\mathrm{ref}} = 60\%$ and $\phi = 1$ for glass–backsheet modules. Cooling lowers the Arrhenius factor but raises RH_{e}; (8) resolves this trade-off. PID acts while the array is energized ($G > 50\,\mathrm{W/m^2}$):

$$R_{\mathrm{PID}} = R_{\mathrm{PID}}^{\mathrm{ref}} (1 - \mu_{\mathrm{PID}}) \langle a_{E_{\mathrm{PID}}}(T_m)\, f_{\mathrm{leak}} \rangle_{\mathrm{on}}, \quad (9)$$

$$f_{\mathrm{leak}} = \left[1 + \exp\left(-\frac{RH - RH_{50}^{*}}{w} \right) \right]^{-1}, \qquad RH_{50}^{*} = RH_{50} - \Delta_s\, s, \quad (10)$$

with $E_{\mathrm{PID}} = 0.9\,\mathrm{eV}$, $RH_{50} = 70\%$, width $w = 5\%$ RH and threshold shift $\Delta_s = 15\%$ RH. The salt coverage is $s = \min(m_s/0.5,\, 1)$ with m_s in $\mathrm{g/m^2}$. The steep humidity dependence of PID leakage [12, 13] and the drop in glass surface resistance with sea salt [14, 15] are experimentally supported. The logistic form and its parameters are modelling choices that have not been fitted to data, so this PID submodel is a physically motivated scenario model, not a calibrated predictive model. Corrosion uses the ISO 9223 variables [16]:

$$R_{\mathrm{C}} = R_{\mathrm{C}}^{\mathrm{ref}}\, \frac{TOW}{TOW_{\mathrm{ref}}} \sqrt{\frac{D_{\mathrm{Cl}}}{D_{\mathrm{ref}}}}\, (1 - \mu_{\mathrm{C}}), \quad (11)$$

where TOW counts hours with ambient $RH > 80\%$ or dew on the module, $TOW_{\mathrm{ref}} = 2500\,\mathrm{h}$ and $D_{\mathrm{ref}} = 60\,\mathrm{mg/(m^2\,d)}$. This is a calibrated scenario representation, not a fitted corrosion-damage function. The reference

Table 2: Mitigation options: qualification standards and model assumptions. "(M)" denotes PID-resistant and salt-mist-hardened modules. Cost adders: 0.5, 1.0 and 2.0 INR/Wp for the three module options; 100 INR/kWp/yr per additional fortnightly cleaning.

Option	Qualification	Model representation	Assumed value (MC range)
PID-resistant cell/encapsulant	IEC TS 62804-1:2025	$(1-\mu_{\mathrm{PID}})$ on R_{PID}	$\mu_{\mathrm{PID}} = 0.9$ (0.5–0.95)
Salt-mist-hardened module	IEC 61701:2020	$(1-\mu_{\mathrm{C}})$ on R_{C}	$\mu_{\mathrm{C}} = 0.5$ (0.2–0.7)
Glass–glass, low-permeability	No specific standard	Factor ϕ on RH_{e}	$\phi = 0.8$
More frequent cleaning	O&M practice	Cleaning interval	7 days (vs. 15)

rates ($R_{\mathrm{TH}}^{\mathrm{ref}} = 0.10$, $R_{\mathrm{PID}}^{\mathrm{ref}} = 0.06$, $R_{\mathrm{C}}^{\mathrm{ref}} = 0.10\%$/yr) were calibrated so that standard modules on land degrade by roughly 0.9–1%/yr, the magnitude reported for hot–humid Indian climates [48, 49].

2.8. Mitigation options

Qualification standards define test procedures; they do not establish a numerical reduction of field degradation. Mitigation therefore enters through explicitly assumed factors (Table 2). Salt-mist resistance is qualified with IEC 61701 [50], and PID susceptibility with IEC TS 62804-1 [51].

2.9. Lifetime energy and economics

Year-y energy is $E_y = E_1(1 - R/100)^{y-1}$ over $N = 25$ years. This assumes a constant effective annual rate derived from the simulated year's stress distribution. Consequently, the lifetime estimates are equivalent constant-rate projections, not predictions of the exact annual degradation trajectory. The levelized cost of energy is

$$\mathrm{LCOE} = \frac{C_0 + \sum_{y=1}^{N} (O_y - B_y)(1+r)^{-y}}{\sum_{y=1}^{N} E_y (1+r)^{-y}}, \tag{12}$$

with discount rate $r = 9\%$, O&M costs O_y escalating by 5%/yr, and an optional water benefit $B_y = c_w V_w$ for the evaporation avoided by a fresh-

water plant ($10\,m^2$/kWp, 65% suppression of 1800 mm/yr). Costs are illustrative Indian-market values, so the economic results are location- and year-dependent.

2.10. Implementation and computational cost

The model is implemented in Python (NumPy, pandas) as an open-source package. All sub-models are vectorized over the 8,760 hours except the three recursive state updates (soiling, dew hysteresis and encapsulant moisture). One scenario-year takes about 23 ms on one core of the test machine. Monte Carlo and Sobol samples are distributed over 4 processes. The parameter vectors are drawn in a fixed order, so serial and parallel runs are bit-identical. The 61,440 scenario evaluations of the Sobol analysis take about 5.1 min on 4 cores.

3. Verification and validation

The V&V assessment follows the established distinction between conceptual model validity, computerized model verification, operational validity and data validity [31, 32].

3.1. Conceptual model validity: evidential basis of the components

Table 3 classifies every model component as an established model used as published (A), a literature-informed parameter value (B), or a modelling assumption introduced here (C). The contribution lies in the coupling and in category C, which is also where field calibration is most needed. Every category-C parameter is varied in the uncertainty analysis (Appendix Appendix A).

3.2. Computerized model verification

Two techniques are used. First, *code-to-code verification*: each category-A sub-model is compared hour by hour, over the full baseline year, with the reference implementation in pvlib [37, 38] (version 0.16.1). Solar position is compared with the NREL solar position algorithm (SPA) [52] (Table 4). Of the 16 comparisons, 13 agree to within 3×10^{-12}, that is, to floating-point

Table 3: Evidential basis of the model components.

	Components
A	*Established models:* Spencer solar position; Haurwitz clear sky; Erbs decomposition; Hay–Davies transposition; ASHRAE IAM; Kasten–Young air mass; Gueymard precipitable water; First Solar spectral factor; Magnus relation; Faiman module temperature; PVWatts DC and inverter models; Arrhenius/Peck functional form; ISO 9223 concepts of time of wetness and chloride deposition; discounted LCOE.
B	*Literature-informed parameters:* land and FPV heat-loss coefficients and their ranges; Peck E_a and n; PID activation energy; NaCl deliquescence RH (75%); ISO 9223 chloride classes; dust loss coefficient; degradation reference rates calibrated to the magnitude of Indian field surveys.
C	*New modelling assumptions:* over-water microclimate (α, ΔT_d, k_v, T_w); module-surface RH with a 48-h encapsulant lag; dew criterion and dew optical loss; wind-proportional salt deposition and deliquescence cementation; salt-assisted logistic leakage function and Δ_s; corrosion scaling $TOW \cdot \sqrt{D_{\mathrm{Cl}}}$; additive mechanisms with a constant annual rate; mitigation factors and cost adders; all deposition rates and costs.

precision. The Erbs decomposition agrees to the same precision when both codes use the same solar constant. With the 1361 W/m^2 used here instead of pvlib's default of 1366.1 W/m^2, the diffuse irradiance differs by at most 4.2 W/m^2. The only genuine approximation is the Spencer solar-position series. Its mean absolute zenith error relative to SPA is 0.09° (maximum 0.25°), and it changes the annual plane-of-array irradiation by 0.02%, which is negligible for this study.

Second, *unit and regression tests* (23 automated tests) check physical limits and invariants:

- the noon zenith at the equinox;
- conservation of GHI in the decomposition;
- a spectral factor of unity at reference conditions;
- round-trip consistency of the dew point;
- PVWatts output at STC and inverter clipping;
- soiling saturation without cleaning;
- monotonicity of the degradation rates in temperature, humidity and chloride;

Table 4: Code-to-code verification against pvlib 0.16.1 over the baseline year (daytime hours unless stated). MAE: mean absolute error; Max: maximum absolute error.

Sub-model (this code vs. pvlib)	Unit	n	MAE	Max
Solar zenith (Spencer vs. SPA)	deg	4,120	0.0905	0.249
Solar azimuth (Spencer vs. SPA)	deg	4,120	0.242	1.73
Extraterrestrial irradiance	%	8,760	5.1×10^{-16}	1.7×10^{-14}
Erbs DHI (common solar constant)	W/m^2	4,119	4.9×10^{-14}	2.0×10^{-12}
Erbs DNI (common solar constant)	W/m^2	4,119	7.3×10^{-14}	2.6×10^{-12}
Erbs DHI (1361 vs. 1366.1 W/m^2)	W/m^2	4,119	1.81	4.19
Hay-Davies sky diffuse	W/m^2	4,119	1.6×10^{-14}	1.1×10^{-13}
Angle of incidence	deg	4,119	1.5×10^{-15}	5.2×10^{-14}
ASHRAE IAM	–	3,739	3.6×10^{-18}	6.7×10^{-16}
Ground-reflected	W/m^2	4,119	0	0
Air mass (Kasten-Young)	%	4,119	1.8×10^{-16}	3.0×10^{-14}
Precipitable water (Gueymard)	%	8,760	0	0
Spectral factor (First Solar)	–	4,119	3.8×10^{-19}	2.2×10^{-16}
Faiman module temperature	K	4,119	0	0
PVWatts DC power	kW	4,119	1.0×10^{-14}	2.3×10^{-13}
PVWatts inverter	kW	4,119	0	0

- reproducibility and serial/parallel equivalence of the Monte Carlo.

3.3. Data validity and the weather generator

Table 5 compares 20 generated years with the climatology they are built from. The generator is climatologically consistent with its inputs; this is a consistency check, not an external validation. The RH match is exact by construction, and the clearness index (hence GHI) is a calibrated input. The check also served as a verification step: an earlier version of the generator over-estimated mean wind by about 0.5 m/s and mean RH by about 2.6 points, and both biases were corrected.

3.4. Operational validity: status and protocol

No measured hourly weather series and no FPV module-temperature, soiling or degradation data were available, so operational validity could not be established. The results are therefore presented as model-based scenario predictions. Table 6 proposes the minimum measurement set for an operating FPV plant with a co-located land reference. It maps each measurement

Table 5: Synthetic-weather consistency check: 20 generated years versus the input climatology. MBE, RMSE and r over the 12 monthly means. †Annual totals; MBE and RMSE in mm/month.

Variable	Normal	Generated	MBE	RMSE	r
Mean air temperature (°C)	28.3	28.1	−0.24	0.36	0.996
Diurnal range (K)	7.4	7.2	−0.20	0.23	0.993
Mean RH (%)	74.0	74.0	0.00	0.00	1.000
Mean wind speed (m/s)	3.3	3.3	0.01	0.03	0.999
Rainfall (mm)†	1,005.0	944.2	−5.07	9.71	0.995

Table 6: Proposed field-validation measurements and the parameters they calibrate.

Measurement	Interval	Calibrates
GHI and POA irradiance (pyranometers)	1 min	Decomposition, transposition, clearness
Air temperature, RH, wind over water and over land	1 min	α, ΔT_d, k_v
Water surface temperature	10 min	T_w
Back-of-module temperature (several positions)	1 min	U_0, U_1
Dew/leaf-wetness sensor on module glass	10 min	Dew model, TOW
Soiling reference cells (cleaned/uncleaned)	1 min	r_d, r_s, k_d, k_s, cementation
Chloride deposition (wet candle, ISO 9225)	Monthly	D_{Cl}
Rainfall; DC power of monitored strings	1 min	Rain cleaning; PR and loss chain
Insulation resistance and leakage current	Daily	f_{leak}, PID onset
IV curves of reference modules	Quarterly	$R_{\mathrm{TH}}^{\mathrm{ref}}$, $R_{\mathrm{PID}}^{\mathrm{ref}}$, $R_{\mathrm{C}}^{\mathrm{ref}}$

to the parameters it would calibrate and prioritizes the inputs that the sensitivity analysis identifies as most influential (Section 5.4).

4. Design of the simulation experiments

4.1. Case study

The reference site is Visakhapatnam (17.69° N, 83.22° E) on the Bay of Bengal. Its baseline synthetic year has a GHI of 1,889 kW h/m^2, a mean air temperature of 28.1 °C, a mean RH of 74.0% and 1,043 mm of rain (Fig. 2). Three 1-MWp plants (DC/AC ratio 1.2, mono-PERC) are compared (Table 7): coastal land PV; freshwater FPV on an inland reservoir;

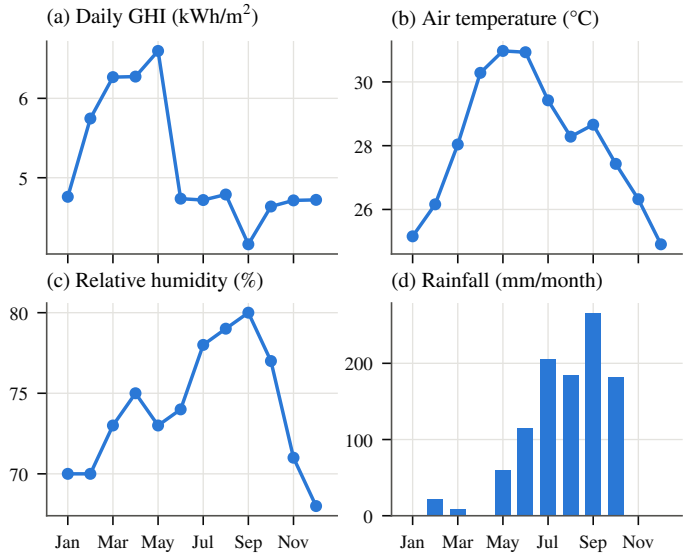


Figure 2: Monthly climate of the baseline synthetic year: (a) daily GHI, (b) air temperature, (c) relative humidity and (d) rainfall.

Table 7: Baseline scenario parameters (assumptions).

Parameter	Land	FPV fresh	FPV saline
Tilt (°); albedo	15; 0.20	12; 0.07	12; 0.07
U_0 (W/(m^2 K)); U_1 (W s/(m^3 K))	25.0; 6.84	35.0; 8.0	35.0; 8.0
Air–water coupling α; dew-point rise ΔT_d (K); wind factor k_v	0; 0; 1.00	0.25; 1.0; 1.15	0.30; 1.5; 1.25
Dust; salt deposition (g/(m^2 d))	0.15; 0.015	0.04; 0.010	0.03; 0.060
D_{Cl} (mg/(m^2 d))	30	8	180
Wiring loss; wave-motion mismatch; availability (%)	1.5; 0; 99.0	2.0; 0.7; 98.5	2.0; 1.0; 98.0
CAPEX (INR/Wp); O&M (INR/kWp/yr)	35; 450	45; 550	52; 750

and saline FPV on near-shore or brackish water. The baseline values are scenario assumptions.

4.2. Experimental design

Table 8 lists the simulation experiments. Two sources of uncertainty are treated separately. *Weather variability* is represented by 20 independent synthetic years (E2). *Parameter uncertainty* is propagated by Monte Carlo sampling of 28 parameters from uniform or log-uniform distributions (E6). The parameters, bounds and justifications are listed in Appendix Appendix A. Parameters shared by the plants take the same draw within a sample, so

Table 8: Design of the simulation experiments. Each run evaluates the scenarios listed.

ID	Experiment	Factors varied	Runs
E1	Baseline	3 plants, standard and mitigated modules	6
E2	Weather variability	20 weather seeds	20×3
E3	Humidity × salinity grid (saline)	ΔT_d (6 levels) × salt/chloride (5)	30
E4	Threshold sweeps	salt (10), chloride (13), ΔT_d fresh (13)	36
E5	Mitigation matrix (saline)	6 option sets	6
E6	Monte Carlo	28 parameters; 1,000 (5,000) samples	4×5,000
E7	Sobol decomposition	28 parameters; $N = 512$	4×15,360
E8	Financing sensitivity	discount rate, O&M escalation	6

scenario differences are paired.

Convergence is assessed by running 5,000 samples with the same seed and evaluating the statistics on nested subsets of 250–5,000 samples, with bootstrap standard errors. The first 1,000 samples are exactly those reported. Spearman rank correlations screen the parameters. First-order and total-order Sobol indices [33] are then estimated from Saltelli sampling [34] with base sample size $N = 512$, that is $N(D + 2) = 15{,}360$ parameter vectors for $D = 28$, using SALib [53, 54] (E7). Experiments E3–E5 and E8 are deterministic sweeps that produce thresholds and break-even values.

5. Results

Unless stated otherwise, results are model predictions with the baseline parameters.

5.1. Thermal behavior and energy yield

The baseline modelled cooling of the floating modules is 3.0 K in daytime, with a peak temperature 7.3 K lower than on land (Fig. 3, Table 9). This follows from the assumed FPV heat-loss coefficients and is not a measured effect. At baseline, freshwater FPV has the highest year-1 yield, 1634 kWh/kWp (+1.6% against land, PR 0.837 against 0.819). The temperature loss falls from 6.12% to 4.78% and the soiling loss from 3.67% to 1.27%.

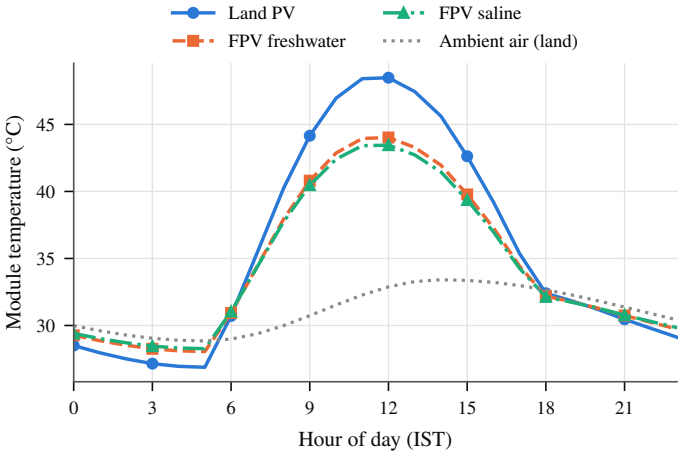


Figure 3: Mean module temperature by hour of day in May (baseline parameters).

Table 9: Year-1 performance (baseline).

Metric	Land	FPV fresh	FPV saline
Specific yield (kWh/kWp)	1608	1634	1590
Performance ratio	0.819	0.837	0.814
Mean daytime / maximum T_m (°C)	39.5 / 57.3	36.5 / 50.0	36.2 / 49.4
Mean air RH at array (%)	74.0	77.5	79.7
Dew on glass; time of wetness (h/yr)	1,163; 3,055	1,092; 3,719	1,389; 4,288
Mean daytime soiling ratio	0.965	0.988	0.969

Part of this gain is lost to higher DC losses, lower availability and slightly lower POA irradiation (Table 10). Salt soiling brings the saline plant's soiling loss back to 3.29%, and its baseline year-1 yield is 1590 kWh/kWp (−1.1%). The absolute soiling losses are scenario estimates, because deposition rates were not measured. Varying the assumed dew optical loss over 0–6% changes the yield differences by less than 0.05 percentage point.

5.2. Module degradation and lifetime energy

Table 11 and Fig. 4 resolve the modelled degradation by mechanism. Floating modules are cooler, but their encapsulant equilibrates at a higher moisture level: mean RH_e is 63.5%, 68.4% and 70.4% at the three plants. Through the cubic humidity term of (8), this cancels the Arrhenius benefit: R_{TH} is 0.34, 0.34 and 0.36%/yr. In this model, cooling by water therefore

Table 10: Year-1 loss breakdown (baseline). Percentages are sequential losses relative to the preceding stage and are not directly additive.

Loss stage (%)	Land	FPV fresh	FPV saline
Incidence angle (IAM)	3.44	3.47	3.47
Spectral (humidity)	-0.23	-0.24	-0.25
Soiling (dust + salt)	3.67	1.27	3.29
Dew / condensation	0.02	0.04	0.05
Module temperature	6.12	4.78	4.64
DC (wiring, mismatch, motion, LID)	3.46	4.62	4.91
Inverter and clipping	2.03	2.03	2.03
Availability	1.00	1.50	2.00

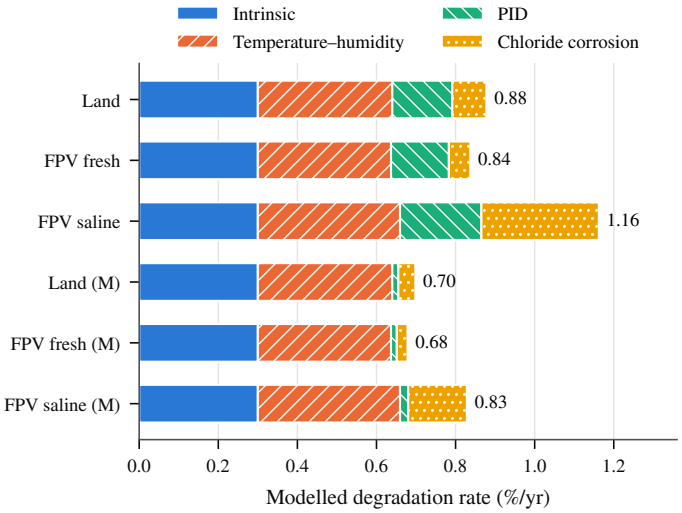


Figure 4: Modelled module degradation rate by mechanism (baseline). (M): mitigated modules with the assumed factors of Table 2.

does not translate into slower moisture-driven ageing at a humid site. PID is highest at the saline plant (0.21%/yr), where the salt film makes the glass conductive at lower RH. Corrosion separates the sites most clearly ($R_C = 0.30$ against 0.09%/yr). Total degradation is 0.88, 0.84 and 1.16%/yr. Over 25 years, the baseline freshwater plant delivers +2.1% and the baseline saline plant −4.3% relative to land (Fig. 5).

Table 11: Modelled module degradation, lifetime energy and LCOE (baseline, per MWp).

	Land	FPV fresh	FPV saline	Land (M)	Fresh (M)	Saline (M)
R_0 intrinsic (%/yr)	0.30	0.30	0.30	0.30	0.30	0.30
R_{TH} (%/yr)	0.34	0.34	0.36	0.34	0.34	0.36
R_{PID} (%/yr)	0.15	0.15	0.21	0.02	0.01	0.02
R_{C} (%/yr)	0.09	0.05	0.30	0.04	0.03	0.15
Total R (%/yr)	**0.88**	**0.84**	**1.16**	**0.70**	**0.68**	**0.83**
Mean encapsulant RH (%)	63.5	68.4	70.4	63.5	68.4	70.4
25-year energy (MWh)	36,233	37,004	34,677	37,002	37,698	36,045
LCOE (INR/kWh)	2.83	3.54	4.43	2.90	3.60	4.43

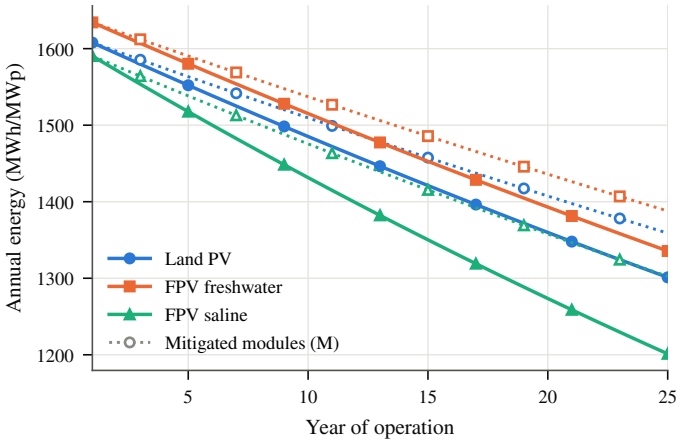


Figure 5: Annual energy over 25 years (baseline). Dotted lines: mitigated modules.

5.3. Thresholds and mitigation

For freshwater FPV, the lifetime advantage over land vanishes at a dew-point rise of $\Delta T_d^* \approx 2.7$ K (Fig. 6b). This is a threshold predicted by the model under the stated assumptions, not a universal constant, but it can serve as a site-screening criterion. Saline FPV out-yields land in year 1 only below about 35 mg/(m^2 d) of salt deposition. Its 25-year energy stays below land at every salt level (-0.9% at zero salt) and at every chloride level in 0–400 mg/(m^2 d) (Fig. 6a). The mitigation matrix (Table 12) shows scenario-based, not validated, benefits. PID-resistant and salt-mist-hardened modules together recover 3.9% of lifetime energy. Only when a lower-permeability glass–glass package is added does the saline plant exceed land (1.5%).

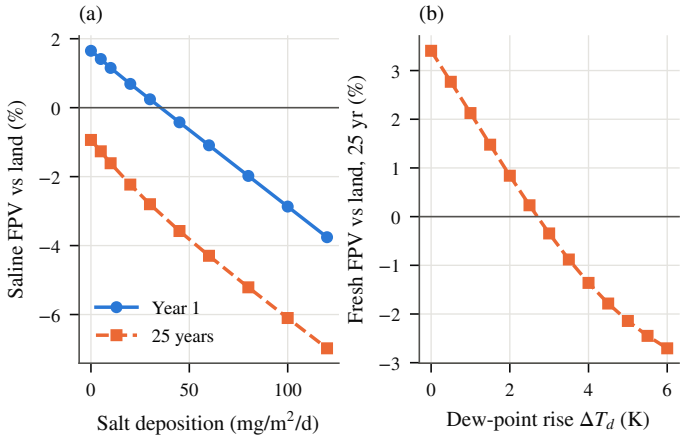


Figure 6: Break-even thresholds relative to land PV. (a) Saline FPV versus salt deposition (chloride at baseline). (b) Freshwater FPV 25-year energy versus dew-point rise over water.

Table 12: Scenario-based mitigation benefits for the saline FPV plant (assumed effects of Table 2; per MWp).

Option	CAPEX (INR/Wp)	R (%/yr)	25-yr energy (MWh)	Recovery (%)	vs. land (%)	LCOE (INR/kWh)
Standard module	52.0	1.16	34,677	0.0	−4.3	4.43
PID-resistant	52.5	0.98	35,427	2.2	−2.2	4.41
Salt-mist-hardened	53.0	1.01	35,277	1.7	−2.6	4.45
PID + salt-mist	53.5	0.83	36,045	3.9	−0.5	4.43
PID + salt-mist + glass-glass	55.5	0.65	36,792	6.1	1.5	4.50
PID + salt-mist + glass-glass + 7-day cleaning	55.5	0.65	37,421	7.9	3.3	4.54

5.4. Weather variability, parameter uncertainty and convergence

Over 20 synthetic years, the yield of the land plant varies between 1,570 and 1,649 kWh/kWp. The 25-year differences between plants, however, vary with a standard deviation of only about 0.1 percentage point ($+2.17 \pm 0.10\%$ for freshwater and $-4.32 \pm 0.06\%$ for saline FPV, Fig. 7). This is stochastic robustness conditional on the generator, not empirical validation.

Parameter uncertainty is much larger (Table 13, Fig. 8). Baseline values and Monte Carlo medians answer different questions. For freshwater FPV, the baseline 25-year difference is $+2.1\%$, whereas the median is $+1.5\%$ with a 95% interval of $[-3.6, 6.3]\%$, positive in 75% of samples. For saline FPV, the baseline is -4.3% and the median -5.1% ($[-11.5, 0.7]\%$), positive in only 4% of samples. The freshwater benefit is therefore conditional, whereas the

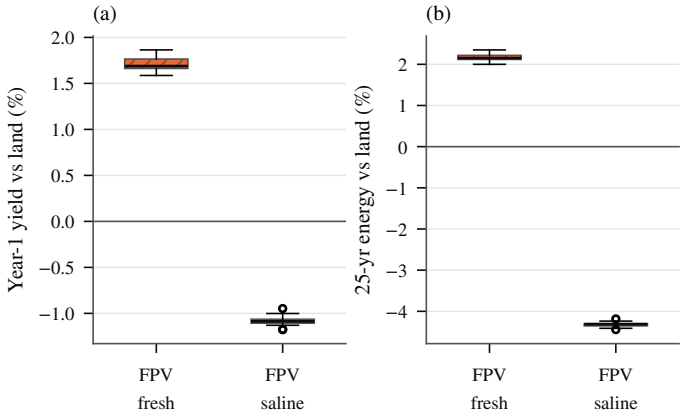


Figure 7: Weather variability only: FPV year-1 yield and 25-year energy relative to land PV over 20 synthetic years.

Table 13: Parameter uncertainty: Monte Carlo over 28 parameters (1,000 samples, baseline weather).

Output	Median	95% interval	$P(>0)$ (%)
Year-1 yield, land (kWh/kWp)	1,604	[1,543, 1,638]	
Year-1 yield, FPV fresh (kWh/kWp)	1,624	[1,589, 1,645]	
Year-1 yield, FPV saline (kWh/kWp)	1,581	[1,520, 1,616]	
Degradation, land (%/yr)	0.89	[0.71, 1.10]	
Degradation, FPV fresh (%/yr)	0.88	[0.68, 1.17]	
Degradation, FPV saline (%/yr)	1.21	[0.91, 1.63]	
Degradation, FPV saline (M) (%/yr)	0.91	[0.67, 1.20]	
Δ yield, fresh − land (%)	1.21	[−1.67, 5.25]	78
Δ yield, saline − land (%)	−1.48	[−5.53, 3.06]	27
Δ 25-yr energy, fresh − land (%)	1.49	[−3.57, 6.33]	75
Δ 25-yr energy, saline − land (%)	−5.08	[−11.50, 0.71]	4
Δ 25-yr energy, saline (M) − land (%)	−1.56	[−6.96, 3.78]	27
LCOE, land (INR/kWh)	2.82	[2.22, 3.58]	
LCOE, FPV fresh (INR/kWh)	3.55	[2.76, 4.55]	
LCOE, FPV saline (INR/kWh)	4.44	[3.52, 5.65]	

saline penalty holds across most of the plausible parameter space. Between 1,000 and 5,000 samples, no median or 95% bound moves by more than 0.28 percentage point, and the bootstrap standard errors at 1,000 samples are at most 0.30 percentage point (Table 14, Fig. 9). 1,000 samples are therefore sufficient.

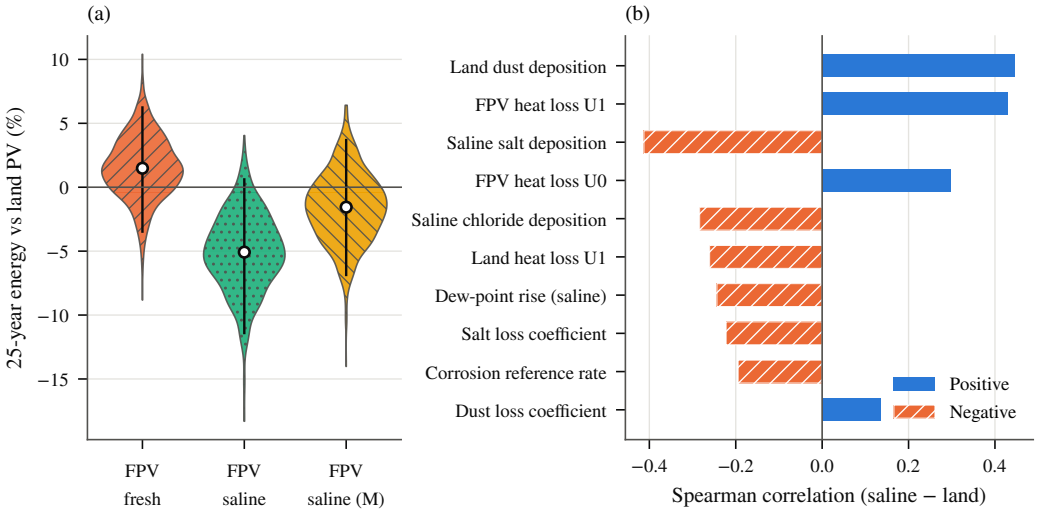


Figure 8: Parameter uncertainty (1,000 samples). (a) 25-year energy relative to land PV (median and 95% interval). (b) Spearman rank correlations with the saline minus land difference.

Table 14: Monte Carlo convergence: 1,000 versus 5,000 samples. ΔE_{25} and ΔE_1: 25-year energy and year-1 yield relative to land PV (%); bootstrap SE of the median in brackets.

Output	Median		95% interval	
	n=1,000	n=5,000	n=1,000	n=5,000
ΔE_{25}, fresh	1.49 (0.09)	1.42	[−3.57, 6.33]	[−3.55, 6.34]
ΔE_{25}, saline	−5.08 (0.11)	−5.12	[−11.50, 0.71]	[−11.22, 0.45]
ΔE_{25}, saline (M)	−1.56 (0.09)	−1.65	[−6.96, 3.78]	[−7.00, 3.65]
ΔE_1, fresh	1.21 (0.07)	1.11	[−1.67, 5.25]	[−1.74, 5.25]

5.5. Variance-based sensitivity

Table 15 and Fig. 10 give the Sobol indices of the 25-year energy differences. For freshwater FPV, Land dust deposition ($S_T = 0.35$), FPV U_1 (0.26) and FPV U_0 (0.19) dominate. For saline FPV, Land dust deposition (0.22), FPV U_1 (0.21) and Saline salt deposition (0.17) lead. The first-order indices sum to 0.94 and 0.97, and the total-order indices to 1.04 and 1.07. The outputs are therefore close to additive in the parameters: interactions explain less than about 10% of the variance. This confirms the Spearman screening and justifies one-at-a-time threshold analyses. Only 4 parameters have S_T significantly above 0.05 for each output. Calibrating the float heat-

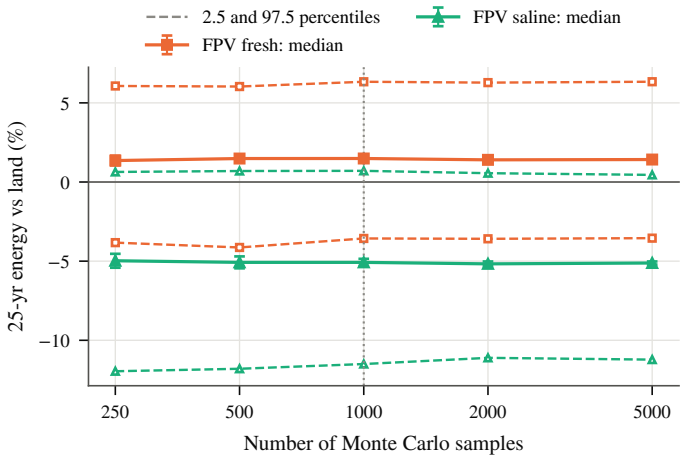


Figure 9: Convergence of the Monte Carlo median (±1.96 bootstrap SE) and 95% bounds on nested subsets of one 5,000-sample run. The dotted line marks the 1,000 samples used.

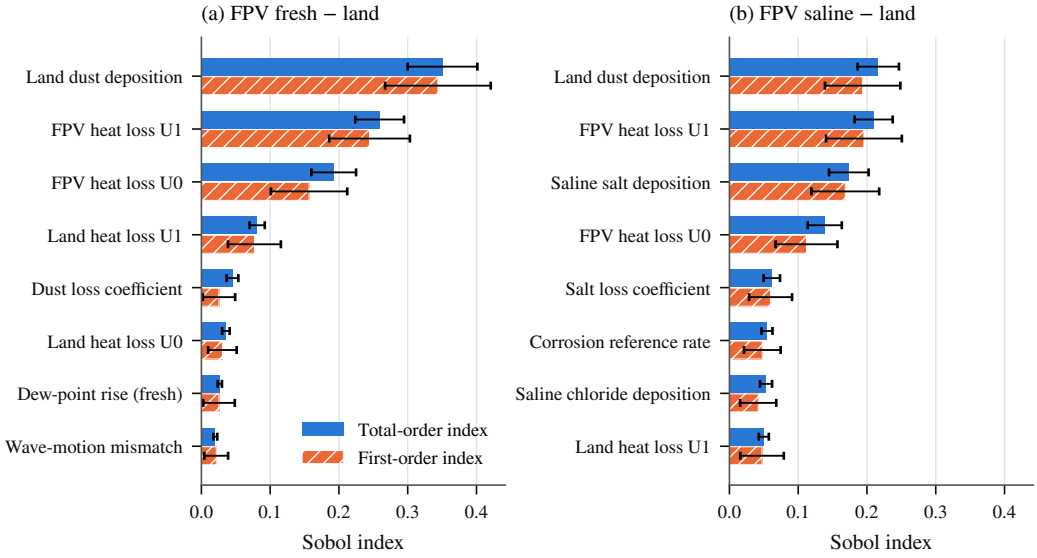


Figure 10: First-order and total-order Sobol indices (bootstrap 95% confidence intervals) of the 25-year energy difference relative to land PV, for the eight most influential parameters: (a) freshwater FPV, (b) saline FPV.

loss coefficients, the dust deposition at the land alternative and, for saline sites, the salt and chloride deposition would remove most of the decision-relevant uncertainty.

Table 15: Sobol indices of the 25-year energy difference relative to land PV ($N = 512$, $\pm$ bootstrap 95% confidence interval), ten most influential parameters.

	FPV fresh − land		FPV saline − land	
Parameter	S_1	S_T	S_1	S_T
Land dust deposition (multiplier)	0.34	0.35±0.05	0.19	0.22±0.03
FPV U_1 (W s/m^{3}K)	0.24	0.26±0.04	0.20	0.21±0.03
FPV U_0 (W/m^{2}K)	0.16	0.19±0.03	0.11	0.14±0.02
Saline salt deposition (multiplier)	0.00	0.00±0.00	0.17	0.17±0.03
Land U_1 (W s/m^{3}K)	0.08	0.08±0.01	0.05	0.05±0.01
Dust loss coefficient k_d (m^2/g)	0.03	0.05±0.01	0.02	0.03±0.01
Salt loss coefficient k_s (m^2/g)	0.00	0.00±0.00	0.06	0.06±0.01
Land U_0 (W/m^{2}K)	0.03	0.04±0.01	0.02	0.02±0.00
Corrosion reference rate R_C^{ref} (%/yr)	0.00	0.00±0.00	0.05	0.05±0.01
Saline chloride deposition (multiplier)	0.00	0.00±0.00	0.04	0.05±0.01

Table 16: LCOE sensitivity to financing assumptions (INR/kWh, baseline energy).

Case	Land	FPV fresh	FPV saline	Fresh − land	Saline − land
Discount rate 6%	2.34	2.92	3.70	0.58	1.36
Discount rate 9%	2.83	3.54	4.43	0.71	1.60
Discount rate 12%	3.37	4.23	5.24	0.86	1.87
O&M escalation 2%	2.72	3.41	4.24	0.69	1.52
O&M escalation 5%	2.83	3.54	4.43	0.71	1.60
O&M escalation 7%	2.94	3.67	4.61	0.73	1.67

5.6. Economics

Under the assumed Indian-market cost and financing conditions, FPV is more expensive per kWh: 3.54 (freshwater) and 4.43 INR/kWh (saline), against 2.83 INR/kWh on land. This holds in 99.5% and 100.0% of Monte Carlo samples. Parity with land PV requires a CAPEX of 34.3 (freshwater; baseline 45) or 29.1 INR/Wp (saline; baseline 52). Alternatively, land PV would need a further 10.5 or 23.6 INR/Wp of land-related cost. Valuing the avoided evaporation (11,700 m^3 per MWp and year) at 10–40 INR/m^3 lowers the freshwater LCOE to 3.47–3.24 INR/kWh, and parity needs about 93 INR/m^3. Financing assumptions shift all LCOEs but not their ranking (Table 16).

6. Discussion

6.1. Simulation-methodological lessons

Three lessons apply beyond this case:

- *Coupling through a shared state.* The coupling through a shared microclimate and module state is essential. Evaluating the thermal benefit and the humidity penalty with independent inputs would miss the cancellation in R_{TH}, because the variable that drives the benefit (T_m) also drives the penalty (through RH_{s}).

- *Separating the sources of uncertainty.* Weather variability alone suggested a precise freshwater advantage (± 0.1 percentage point). Parameter uncertainty showed that this advantage is conditional. Reporting only weather variability would have overstated the confidence in the result by more than an order of magnitude.

- *Using the sensitivity analysis to design validation.* The near-additive Sobol structure, with few influential parameters, means that a compact measurement campaign (Table 6) would reduce most of the decision-relevant uncertainty. Category-C assumptions that turned out to be insensitive, such as the dew optical loss, need not be measured first.

6.2. Comparison with previous work

The baseline modelled thermal advantage (about 3 K in daytime, 7 K at peak) lies within the range reported for open, well-ventilated float designs [7, 8]. It exceeds that of less ventilated designs [18, 19], which the Monte Carlo covers. The salt-driven increase in degradation is larger than the 0.05–0.13 percentage-point effect reported for offshore modules [29]. The difference arises because salt soiling, salt-assisted PID and corrosion are all included here. Consistent with earlier findings that degradation coefficients must be calibrated to the environment [28, 25], the reference rates used here remain the main calibration need.

6.3. Limitations

- Operational validity has not been established: no measured weather or FPV data were available.
- The over-water microclimate, the deposition rates, the degradation reference rates and the mitigation factors are assumptions. Their uncertainty is propagated, but the ranges themselves are judgements.
- Degradation is linear in time and additive across mechanisms. PID saturation and recovery, interactions between mechanisms, abrupt interconnect failures and the higher operating temperature of glass–glass modules are not modelled.
- Floating-structure degradation is outside the model.
- The economics are specific to the assumed Indian-market conditions.

7. Conclusions

A coupled, multi-time-scale simulation framework for the lifetime performance of floating PV in humid and saline environments has been developed, verified against the pvlib reference library, and analyzed with separated weather and parameter uncertainty. For a hot, humid, coastal site in India, the baseline model predicts a modelled cooling of 3.0 K. This gives freshwater FPV a baseline +1.6% year-1 yield and +2.1% 25-year energy relative to land. The humidity increase cancels the temperature benefit for moisture-driven degradation, and at the saline site salt soiling, salt-assisted PID and chloride corrosion lead to a baseline lifetime deficit of −4.3%.

Under parameter uncertainty, the freshwater benefit is conditional (median +1.5%, positive in 75% of samples). In this model it vanishes above a dew-point rise of about 2.7 K. The saline penalty is robust (median −5.1%). Sobol analysis shows near-additive behavior dominated by the float heat-loss coefficients, the dust load at the land alternative and the salt deposition. These are the quantities a field campaign should measure first.

All results are model-based scenario predictions that require field calibration, for which a measurement protocol is proposed. The open, verified implementation is intended as a reusable simulation tool for siting and design studies of floating PV in other humid and saline regions.

CRediT authorship contribution statement

Subrahmanyam Tanala: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The simulation code (`fpv_analysis`), its unit tests, all input parameters and Monte Carlo ranges in machine-readable form, the random seeds, the pinned package versions and every generated table are openly available at `https://github.com/subrahmanyamtanala-rgb/floating-solar-photovoltaic-system-for-humid-and-saline-conditions`. The command `./reproduce.sh` regenerates every number, table and figure of this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to develop the simulation code, run the simulation experiments, and draft and revise the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A. Monte Carlo parameter space

Table A.1 lists the 28 sampled parameters. "U" and "log-U" denote uniform and log-uniform distributions; multipliers scale the baseline deposition rates of Table 7.

References

[1] World Bank Group, ESMAP, SERIS, Where sun meets water: Floating solar market report, Tech. rep., World Bank, Washington, DC, USA (2019).

[2] A. Sahu, N. Yadav, K. Sudhakar, Floating photovoltaic power plant: A review, Renew. Sustain. Energy Rev. 66 (2016) 815–824.

[3] K. Trapani, M. Redón Santafé, A review of floating photovoltaic installations: 2007–2013, Prog. Photovolt.: Res. Appl. 23 (4) (2015) 524–532.

[4] S. Oliveira-Pinto, J. Stokkermans, Assessment of the potential of different floating solar technologies – overview and analysis of different case studies, Energy Convers. Manage. 211 (2020) 112747.

[5] M. Kumar, et al., Challenges and opportunities towards the development of floating photovoltaic systems, Sol. Energy Mater. Sol. Cells (2021).

Table A.1: Monte Carlo parameters: baseline, distribution, bounds and justification.

Parameter	Baseline	Distribution	Lower	Upper	Justification
Peck reference rate R_{TH}^{ref} (%/yr)	0.1	U	0.06	0.14	Calibration constant; land total ≈0.9–1%/yr [48]; ±40%
PID reference rate R_{PID}^{ref} (%/yr)	0.06	U	0.03	0.09	Calibration constant; ±50%
Corrosion reference rate R_{C}^{ref} (%/yr)	0.1	U	0.05	0.15	Calibration constant; ±50%
Dew optical loss	0.04	U	0	0.06	Assumption; effect shown negligible (Sec. 5.2)
Dust loss coefficient k_d (m^2/g)	0.035	U	0.025	0.045	Order of magnitude of dust-soiling studies [55]
Salt loss coefficient k_s (m^2/g)	0.06	U	0.03	0.09	Assumption; no measured salt-crust transmittance
Deliquescence cementation factor	0.5	U	0.25	0.75	Assumption; salt crusts observed on coastal FPV [30]
Land U_0 (W/m^2K)	25	U	22	29	Around open-rack Faiman defaults [46]
Land U_1 (W s/m^3K)	6.84	U	4.5	8	Around open-rack Faiman defaults [46]
Land dust deposition (multiplier)	1	log-U	0.5	2	Assumption; no site measurement
Land chloride deposition (multiplier)	1	log-U	0.5	2	15–60 mg/m^2/day, ISO 9223 class S1 [16]
FPV U_0 (W/m^2K)	35	U	22	45	Measured FPV values [18, 20, 8, 17]
FPV U_1 (W s/m^3K)	8	U	2.5	10	Measured wind terms [21, 18]
Over-water wind factor k_v	1.15	U	1	1.35	Assumption (lower surface roughness)
Wave-motion mismatch	0.007	U	0.003	0.015	Assumption
Air–water coupling α (fresh)	0.25	U	0.1	0.4	Assumption
Dew-point rise ΔT_d (fresh, K)	1	U	0.5	1.5	Assumption; threshold analysis in Sec. 5.3
Air–water coupling α (saline)	0.3	U	0.15	0.45	Assumption
Dew-point rise ΔT_d (saline, K)	1.5	U	1	2	Assumption
Saline salt deposition (multiplier)	1	log-U	0.5	2	Assumption; threshold analysis in Sec. 5.3
Saline chloride deposition (multiplier)	1	log-U	0.5	2	90–360 mg/m^2/day, ISO 9223 classes S2–S3 [16]
PID mitigation μ_{PID}	0.9	U	0.5	0.95	Assumption (Table 2); not implied by IEC TS 62804-1
Salt-mist mitigation μ_C	0.5	U	0.2	0.7	Assumption (Table 2); not implied by IEC 61701
Land CAPEX (INR/Wp)	35	U	30	40	Illustrative Indian-market range
FPV fresh CAPEX (INR/Wp)	45	U	38	52	Illustrative Indian-market range
FPV saline CAPEX (INR/Wp)	52	U	44	60	Illustrative Indian-market range
Discount rate	0.09	U	0.06	0.12	Typical range for utility-scale PV finance
O&M escalation	0.05	U	0.02	0.07	Assumption

[6] A. S. Makhija, et al., Performance and degradation analysis for different solar photovoltaic technologies under hot and humid environment: A review, Prog. Energy (2023).

[7] H. Liu, V. Krishna, J. Lun Leung, T. Reindl, L. Zhao, Field experience and performance analysis of floating PV technologies in the tropics, Prog. Photovolt.: Res. Appl. 26 (12) (2018) 957–967.

[8] M. Dörenkämper, et al., The cooling effect of floating PV in two differ-

ent climate zones: A comparison of field test data from the Netherlands and Singapore, Solar Energy 214 (2021) 239–247.

[9] S. Z. Golroodbari, W. van Sark, Simulation of performance differences between offshore and land-based photovoltaic systems, Prog. Photovolt.: Res. Appl. 28 (9) (2020) 873–886.

[10] D. S. Peck, Comprehensive model for humidity testing correlation, in: Proc. 24th Int. Rel. Phys. Symp. (IRPS), 1986, pp. 44–50.

[11] M. D. Kempe, Modeling of rates of moisture ingress into photovoltaic modules, Sol. Energy Mater. Sol. Cells 90 (16) (2006) 2720–2738.

[12] S. Hoffmann, M. Koehl, Effect of humidity and temperature on the potential-induced degradation, Prog. Photovolt.: Res. Appl. (2014).

[13] W. Luo, et al., Potential-induced degradation in photovoltaic modules: A critical review, Energy Environ. Sci. (2017).

[14] P. Hacke, et al., Effects of photovoltaic module soiling on glass surface resistance and potential-induced degradation, in: Proc. 42nd IEEE Photovolt. Spec. Conf. (PVSC), 2015.

[15] M. A. Islam, et al., Effect of different factors on the leakage current behavior of silicon photovoltaic modules at high voltage stress, IEEE J. Photovolt. (2018).

[16] Corrosion of metals and alloys—corrosivity of atmospheres—classification, determination and estimation, ISO 9223:2012 (2012).

[17] D. Lindholm, et al., Heat loss coefficients computed for floating PV modules, Prog. Photovolt.: Res. Appl. (2021).

[18] V. Nysted, et al., Floating photovoltaics: Modelled and experimental operating temperatures and the impact of wind speed and direction, EPJ Photovolt. (2024).

[19] C. Fırat, Thermal performance and temperature correction of floating photovoltaic arrays based on high-resolution field measurements, Int. J. Ambient Energy (2026).

[20] R. Wu, et al., Heat dissipation effect of the floating photovoltaic for an inland water test site in China, Energy Convers. Manage. (2024).

[21] M. Dörenkämper, et al., The impact of system sizing and water temperature on the thermal characteristics of floating photovoltaic systems, Energies (2024).

[22] D. Thevenard, S. Pelland, Estimating the uncertainty in long-term photovoltaic yield predictions, Solar Energy (2013).

[23] M. Theristis, et al., Blind photovoltaic modeling intercomparison: A multidimensional data analysis and lessons learned, Prog. Photovolt.: Res. Appl. (2023).

[24] J. J. Roberts, et al., Assessment of photovoltaic performance models for system simulation, Renew. Sustain. Energy Rev. (2017).

[25] M. A. Hameed, et al., Analysis and prediction of the performance and reliability of PV modules installed in harsh climates: Case study Iraq, Renew. Energy (2024).

[26] I. Kaaya, et al., Modeling outdoor service lifetime prediction of PV modules: Effects of combined climatic stressors on PV module power degradation, IEEE J. Photovolt. 9 (4) (2019) 1105–1112.

[27] D. Yuniawan, et al., Stochastic–dynamic modelling of performance degradation in a grid-connected rooftop photovoltaic system, J. Eng. Technol. Sci. (2026).

[28] G. Mannino, et al., Photovoltaic module degradation forecast models for onshore and offshore floating systems, Energies (2023).

[29] A. Ajmal, et al., Degradation rate analysis of offshore PV module applications considering climatic stresses and salinity effects, Solar Energy (2025).

[30] A. Ahmad, et al., Impact of coastal extreme weather stressors on bifacial floating-based and ground solar PV systems: Thermal performance, salt crystallization and dust accumulation, Therm. Sci. Eng. Prog. (2026).

[31] R. G. Sargent, Verification and validation of simulation models, J. Simul. 7 (1) (2013) 12–24.

[32] W. L. Oberkampf, C. J. Roy, Verification and Validation in Scientific Computing, Cambridge University Press, Cambridge, 2010.

[33] I. M. Sobol', Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates, Math. Comput. Simul. 55 (1–3) (2001) 271–280.

[34] A. Saltelli, P. Annoni, I. Azzini, F. Campolongo, M. Ratto, S. Tarantola, Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index, Comput. Phys. Commun. 181 (2) (2010) 259–270.

[35] Y. Xiong, et al., Mechanical properties of offshore floating photovoltaic structural coating/carbon steel system in a marine environment, Ocean Eng. (2025).

[36] K. Feng, et al., Mechanical performance of key components in floating photovoltaic systems: Technological advances and application prospects, Global Energy Interconnection (2025).

[37] W. F. Holmgren, C. W. Hansen, M. A. Mikofski, pvlib python: A python package for modeling solar energy systems, J. Open Source Softw. 3 (29) (2018) 884.

[38] K. S. Anderson, et al., pvlib python: 2023 project update, J. Open Source Softw. 8 (92) (2023) 5994.

[39] B. Haurwitz, Insolation in relation to cloudiness and cloud density, J. Meteorol. 2 (1945) 154–166.

[40] J. W. Spencer, Fourier series representation of the position of the sun, Search 2 (5) (1971) 172.

[41] D. G. Erbs, S. A. Klein, J. A. Duffie, Estimation of the diffuse radiation fraction for hourly, daily and monthly-average global radiation, Solar Energy 28 (4) (1982) 293–302.

[42] J. E. Hay, J. A. Davies, Calculation of the solar radiation incident on an inclined surface, in: Proc. 1st Canadian Solar Radiation Data Workshop, 1980, pp. 59–72.

[43] M. Lee, A. Panchula, Spectral correction for photovoltaic module performance based on air mass and precipitable water, in: Proc. 43rd IEEE Photovolt. Spec. Conf. (PVSC), 2016, pp. 1351–1356.

[44] F. Kasten, A. T. Young, Revised optical air mass tables and approximation formula, Appl. Opt. 28 (22) (1989) 4735–4738.

[45] C. Gueymard, Analysis of monthly average atmospheric precipitable water and turbidity in Canada and northern United States, Solar Energy 53 (1) (1994) 57–71.

[46] D. Faiman, Assessing the outdoor operating temperature of photovoltaic modules, Prog. Photovolt.: Res. Appl. 16 (4) (2008) 307–315.

[47] A. P. Dobos, PVWatts version 5 manual, Tech. Rep. NREL/TP-6A20-62641, National Renewable Energy Laboratory, Golden, CO, USA (2014).

[48] R. Dubey, et al., All-india survey of photovoltaic module degradation: 2014, Tech. rep., National Centre for Photovoltaic Research and Education (NCPRE), IIT Bombay, and National Institute of Solar Energy (NISE), Mumbai, India (2015).

[49] D. C. Jordan, S. R. Kurtz, Photovoltaic degradation rates—an analytical review, Prog. Photovolt.: Res. Appl. 21 (1) (2013) 12–29.

[50] Photovoltaic (PV) modules—salt mist corrosion testing, IEC 61701:2020 (2020).

[51] Photovoltaic (PV) modules—test methods for the detection of potential-induced degradation—Part 1: Crystalline silicon, IEC TS 62804-1:2025 (2025).

[52] I. Reda, A. Andreas, Solar position algorithm for solar radiation applications, Solar Energy 76 (5) (2004) 577–589.

[53] J. Herman, W. Usher, SALib: An open-source Python library for sensitivity analysis, J. Open Source Softw. 2 (9) (2017) 97.

[54] A. Saltelli, et al., Global Sensitivity Analysis: The Primer, John Wiley & Sons, Chichester, 2008.

[55] T. Sarver, A. Al-Qaraghuli, L. L. Kazmerski, A comprehensive review of the impact of dust on the use of solar energy: History, investigations, results, literature, and mitigation approaches, Renew. Sustain. Energy Rev. 22 (2013) 698–733.